Secure Energy-Efficient Scheduling for AI Workloads

Placing AI jobs only where they are allowed to run, at the lowest energy that still meets the deadline: a reproducible simulation study and a PhD research programme

github.com/Maxxtra/phd
code · data · every figure

Costin-Alexandru Deonise · Supervisor: Prof. Dr. Eng. Florin Pop

Faculty of Automatic Control and Computers, National University of Science and Technology POLITEHNICA Bucharest · PRECIS Research Centre

KEY FINDINGS Stage 1 · simulated cluster · modelled energy from synthetic profiles · 30 paired seeds · preregistered

7.8 % → 3.7 %
fewer missed deadlines than EFT
Explained by least-slack-first ordering: the EFT-slack ablation reaches the same mean (3.7 %).

−0.6 %
dynamic energy vs EFT
95 % CI [−1.1, −0.1]; H1 supported (Holm $p = 0.040$). Small: lower in 20 of 30 seeds.

≈ 0.2 %
of total energy left for placement
Mean (worst seed 0.7 %), in this model; 36 % is idle power on empty nodes.

1.4 % → 0.2 %
energy gain as secure jobs rise to 80 %
H2 not supported: security constraints shrink the gain instead of growing it.

Why it matters

Schedulers treat *who may see the data* and *what a placement costs in energy* as separate problems.

- AI's energy and environmental footprint is large and under-reported [7, 8, 9].
- Protected execution costs up to ≈ 7 % throughput for LLM inference on confidential H100 GPUs [6], and confidential GPUs are scarce.
- To our knowledge, energy-aware schedulers [1, 2, 4] do not treat security eligibility as a constraint.

Contribution

- **seas**: eligibility first, then energy vs response time; every decision explained.
- A preregistered, reproducible benchmark with ablations.
- Evidence, in this simulated setting, of where placement-only energy optimisation stops paying off.

Question & hypotheses

If security eligibility is a hard constraint, how much *modelled* energy can placement still save without missing more deadlines?

- **H1** **seas** uses less dynamic energy than EFT *and* is non-inferior on misses (margin 2 pp).
- **H2** The gain grows with the share of security-constrained jobs.
- **H3** The energy gain vs EFT and fewer misses than energy-greedy survive noisy estimates ($\sigma = 25\,\%$).

Preregistered; paired tests; Holm correction.

How seas decides

A node is *eligible* only if it offers every required capability and enough memory:

$$S_{jn} = \mathbb{1}\left[R_j \subseteq C_n,\ m_j \le M_n\right]$$

Among eligible nodes that still meet the deadline, **seas** picks

$$n^* = \arg\min_n\ \alpha\,\hat{E}_{jn} + (1-\alpha)\,\hat{R}_{jn}$$

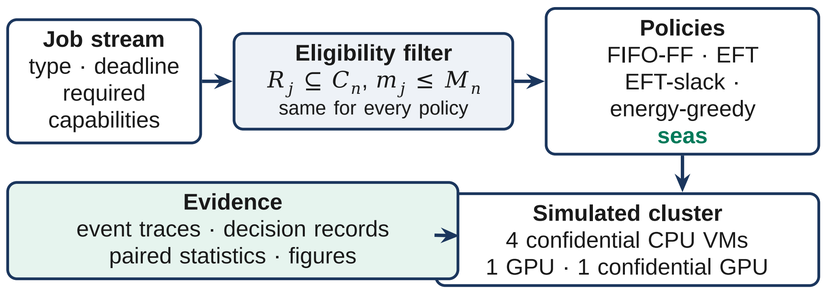
($\hat{E}$ energy, $\hat{R}$ response time, both relative to the best option). Least-slack first, online, non-preemptive; a greedy heuristic.

Job j0006 · infer_small · requires TEE_CPU
deadline in 5.7 s · α = 0.5 · score = α·Ê + (1−α)·R̂

Node	time	energy	decision
gpu-0	1.1 s	136 J	ineligible: no TEE_CPU
cpu-tee-0	25.1 s	997 J	misses deadline
cpu-tee-1	25.1 s	997 J	misses deadline
cpu-tee-2	25.1 s	997 J	misses deadline
cpu-tee-3	25.1 s	997 J	misses deadline
gpu-tee-0	1.1 s	146 J	chosen: lowest score

A logged decision (S1, seed 0). The plain GPU (estimate shown if it ran unprotected) is ineligible; the confidential CPU VMs would miss the deadline; the confidential GPU is chosen.

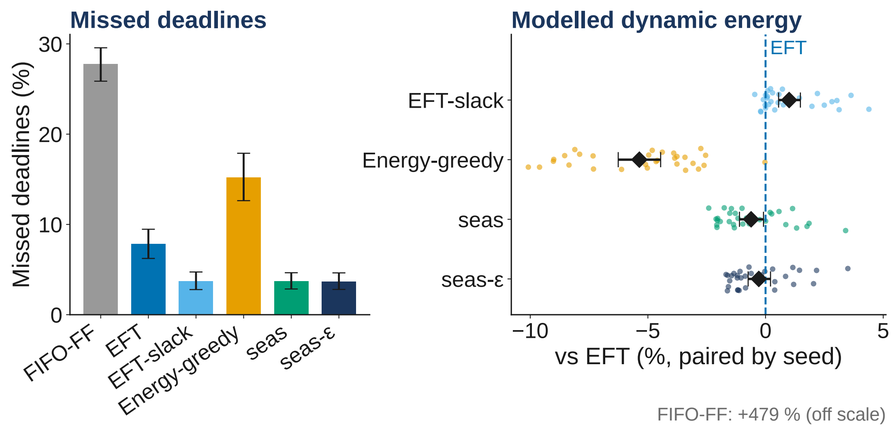
What is built and tested



IMPLEMENTED & TESTED Open-source Python: deterministic discrete-event simulator, 6 policies incl. an ablation, explainable decisions, verification gates, CI that re-runs the confirmatory experiment. **0 eligibility violations** in 4 380 runs, each trace replayed independently.

Security is *simulated* as capability eligibility: no enclave, encryption or attestation is executed [5].

Results: deadlines and energy (H1)



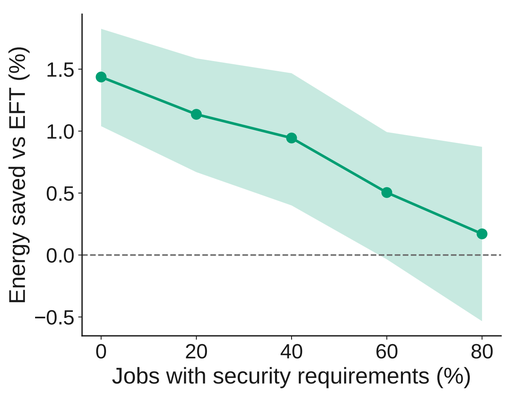
FIFO-FF: +479 % (off scale)

S1, 30 paired seeds. Left: mean miss rate with 95 % bootstrap CI. Right: each dot is one seed; diamond = mean with 95 % CI.

H1 supported (Holm $p = 0.040$): energy −0.6 % [−1.1, −0.1] and misses non-inferior (upper bound −3.2 pp < 2 pp).

Where the gains come from. Ordering jobs by slack (EFT-slack) cuts misses to 3.7 % but costs +1.0 % energy; the energy term in **seas** then saves −1.6 % against that same ordering. Deadline-blind energy-greedy uses −4.8 % energy vs **seas** but misses +11.5 pp more deadlines.

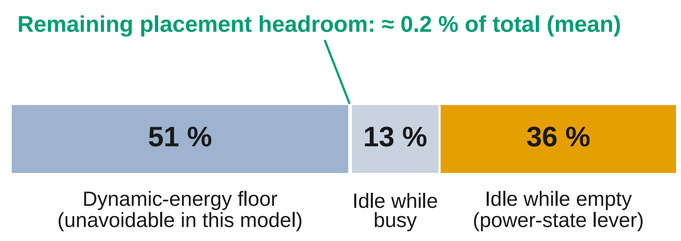
Security shrinks the gain (H2)



H2 not supported.
The gain fell from 1.4 % to 0.2 % (slope −1.6 pp per 100 % secure jobs; post hoc two-sided $p < 0.001$). Interpretation: eligibility removes choices before energy can be optimised.

Energy saved vs EFT, mean and 95 % CI over 30 paired seeds per level (S4).

Limits of placement alone



Exploratory. Where **seas**'s modelled energy goes (S1 mean; always-on nodes over the service window). Floor: relaxation bound, each completed job on its cheapest eligible node.

- Placement only moves dynamic energy, and **seas** is within 0.4 % of its floor (EFT: 2.4 %).
- Under ideal power gating the energy saving disappears (total +0.17 %, CI [−0.24, +0.62]; exploratory).
- Robust to noisy estimates, $\sigma = 25\,\%$ (H3 supported, Holm $p = 0.002$). The weight α barely matters (exploratory): identical schedules for $\alpha \ge 0.25$ in 27 of 30 seeds.

Negative results set the agenda: the remaining energy is in *when nodes are on*, and in whether the model matches hardware.

Next steps

1 · Power-state-aware scheduling
Decide placement together with node sleep/wake-up and frequency, including wake-up costs; compare with an exact bound on small instances.

2 · Hardware energy calibration
Measure CPU (RAPL) and GPU (NVML) power per job type; report modelled-vs-measured error before any energy claim.

PhD roadmap

IMPLEMENTED Stage 1: security-eligible energy-aware scheduler in simulation, benchmark, statistics.

PLANNED Not yet implemented; long-term objectives:

- **2** · Hardware calibration (measured energy)
- **3** · Real TEEs: measured overhead, attestation, recovery
- **4** · Continual-learning retraining as scheduled jobs
- **5** · QPU as a resource class, simulators first [10]

Get involved

Open problems for students and collaborators: power-state policies, exact bounds, energy measurement, TEE benchmarks. One command reproduces every figure.

Contact: deonisealex@gmail.com

[1] Wang et al., DVFS-enabled heterogeneous clusters, *IEEE TPDS* 33(12), 2022.
[2] Chau et al., DVFS for CPU-GPU systems, *ACM e-Energy*, 2017.
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[4] You et al., Zeus, *USENIX NSDI*, 2023.
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[7] Luccioni et al., Power hungry processing, *ACM FAccT*, 2024.
[8] Schwartz et al., Green AI, *CACM* 63(12), 2020.
[9] Aczel et al., Environmental cost of AI's energy use, UNU-INWEH, 2026.
[10] Beck et al., QC in HPC ecosystems, *FGCS* 161, 2024.

Scope. All results are simulated; energy is modelled, not measured. No project grant is associated with this work.